Volume of Revolution – Part 2

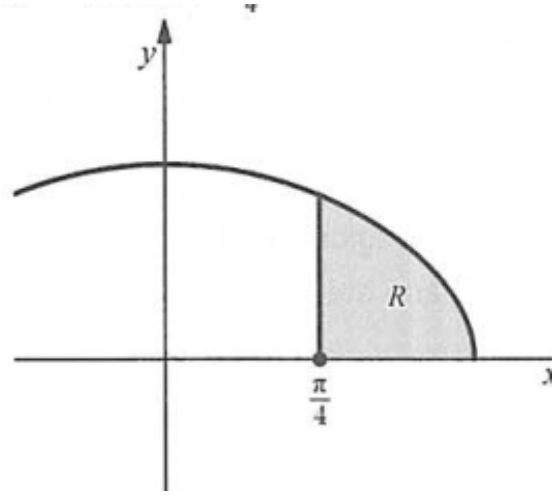
Questions 1–41

Question 1. Find the exact volume created by rotating the curve through 2π radians around the x-axis for each of the following:

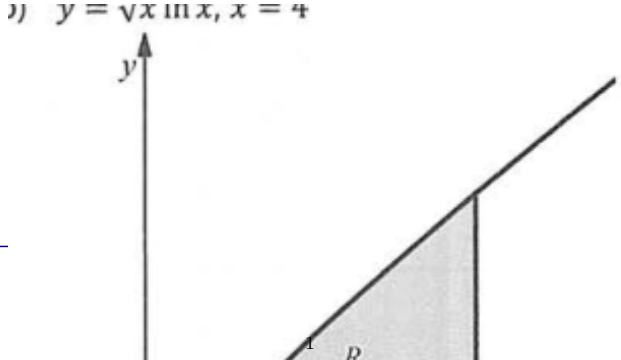
a) $y = (2x-1)^{1/3}$ for $1 \leq x \leq 14$
b) $y = \sec x$ for $\frac{\pi}{6} \leq x \leq \frac{\pi}{3}$
c) $y = \sqrt{\ln x}$ for $1 \leq x \leq 2$
d) $y = \sqrt{\frac{2\cos x}{1+\sin x}}$ for $0 \leq x \leq \frac{\pi}{2}$
e) $y = \sqrt{x}\ \operatorname{cosec} x$ for $\frac{\pi}{4} \leq x \leq \frac{\pi}{2}$
f) $y = \sec x + \tan x$ for $0 \leq x \leq \frac{\pi}{4}$

Question 2. For each of the following, the shaded region is bounded by the given lines, the curve and the x-axis. In each case, find the exact volume of the solid generated when the region is rotated 2π radians about the x-axis:

a) $y = \sqrt{\cos x}$, $x = \frac{\pi}{4}$



b) $y = \sqrt{x}\ln x$, $x = 4$



Question 3. A region R is bounded by the curve $y = \sin x$, the x-axis, the y-axis and the line $x = \pi$. The region is rotated through 2π radians about the x-axis to form a solid of revolution.

a) Sketch the region R.

b) Find the exact volume of the solid formed.

Prerequisite Knowledge for Questions 4–5: Calculus of Hyperbolic Functions

Question 4. A region R is bounded by the curve $y = \sinh x$, the x-axis, the y-axis and the line $x = \ln 5$. The region is rotated through 2π radians about the x-axis to form a solid of revolution.

a) Sketch the region R.

b) Find the exact volume of the solid formed.

Question 5. A region R is bounded by the curve $y = \cosh x$, the x-axis, the y-axis and the line $x = \ln 3$. The region is rotated through 2π radians about the x-axis to form a solid of revolution.

a) Sketch the region R.

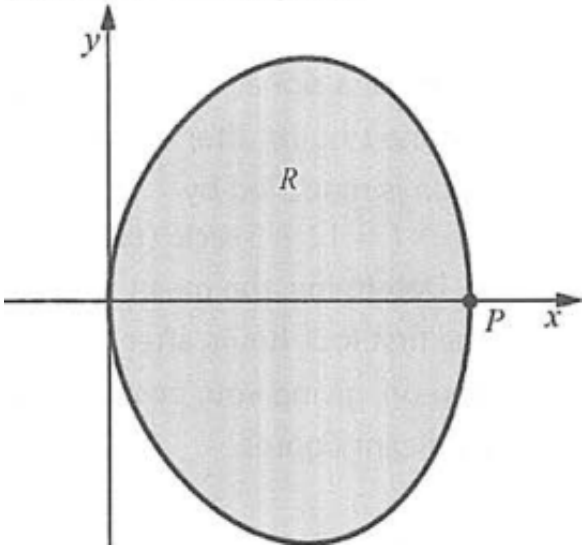
b) Find the exact volume of the solid formed.

Question 6. A shaded region is bounded by the curve $y^2 = 2x\cos x$, the y-axis and the line $x = P$ where P is the point shown in the diagram. The region is rotated around the x-axis until a solid of revolution is generated.

a) Find the value of P.

b) Determine the volume of the solid generated giving your answer in exact form.

t shown in the diagram.



Question 7. A designer is designing a candle holder which is 12 cm long, with flat circular ends. The candle holder is symmetrical about its midpoint and should have the following diameters:

- 5 cm at each end.
- 2 cm at the midpoint.

The shape is modelled by rotating a curve about the x-axis. The origin O is at the centre of the candle holder and the ends are at $x = \pm 6$. Two possible equations for the curve are:

$$\text{Model A: } y = a + kx^2 \qquad \text{Model B: } y = a + m\left(1 - \cos\frac{\pi x}{6}\right)$$

for constants a, k and m.

a) State the value of a which is the radius of the candle holder at the midpoint.

b) Show that $k = \frac{1}{24}$ and $m = \frac{3}{4}$.

c) Use integration to find the volume predicted by both models giving your answer to three significant figures.

d) The designer wanted the model to change gradually near the midpoint and more rapidly towards the ends. Suggest which of the two models is more appropriate.

Prerequisite Knowledge for Questions 8–10: A-Level Maths Partial Fractions

Question 8.

a) Write $\dfrac{6x+11}{(x+3)(2x-1)}$ as partial fractions.

A region R is bounded by the curve $y = \sqrt{\dfrac{6x+11}{(x+3)(2x-1)}}$, the x-axis, the line $x = 1$ and the line $x = 4$. The region is rotated through 2π radians about the x-axis to form a solid of revolution.

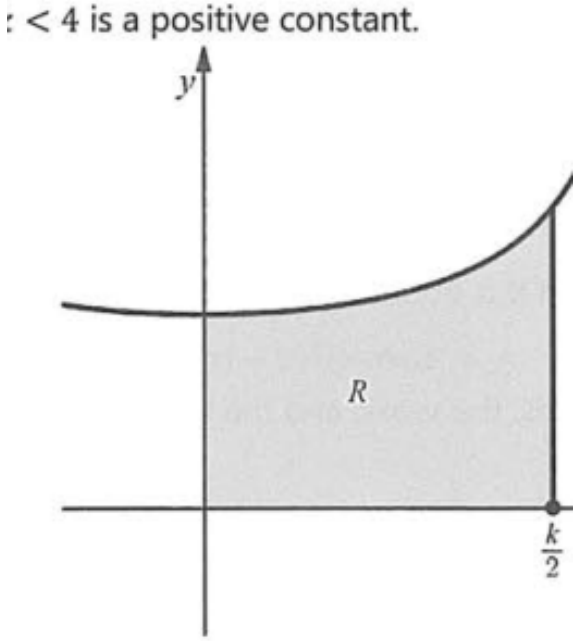
b) Find the exact volume of the solid formed.

Question 9. A region R is bounded by the curve $y = \dfrac{3x+4}{(x+2)\sqrt{x}}$, the x-axis, the line $x = 1$ and the line $x = 3$. The region is rotated through 2π radians about the x-axis to form a solid of revolution. Show that the volume is given by

$$\pi\left(\ln\left(\frac{5^a}{3^b}\right) - \frac{4}{15}\right)$$

where a and b are integers to be found.

Question 10. A shaded region R is bounded by the curve $y = \dfrac{1}{\sqrt{k-x^2}}$, the x-axis, the y-axis and the line $x = \frac{k}{2}$, where $k < 4$ is a positive constant. The region is rotated around the x-axis until a solid of revolution is generated.



a) Show that the exact volume of the solid generated is given by

$$V = \frac{\pi}{2\sqrt{k}}\ln\left(\frac{2\sqrt{k}+k}{2\sqrt{k}-k}\right).$$

b) Given that the region lies entirely between $x = 0$ and $x = \frac{1}{2}$, find the exact volume.

Question 11. A region R is bounded by the curves $y = 1 + \sin x$ and $y = 1 + \cos x$, the line $x = \frac{\pi}{4}$ and the line $x = \frac{5\pi}{4}$. The region is rotated through 2π radians about the x-axis.

a) Sketch the region R.

b) Show that the exact volume of the solid formed is given by $k\sqrt{2}\pi$, where k is a constant to be found.

Prerequisite Knowledge for Question 12: De Moivre's Theorem

Question 12.

a) Use De Moivre's theorem to determine the constants a, b and c such that $\sin^4\theta = a\cos 4\theta + b\cos 2\theta + c$.

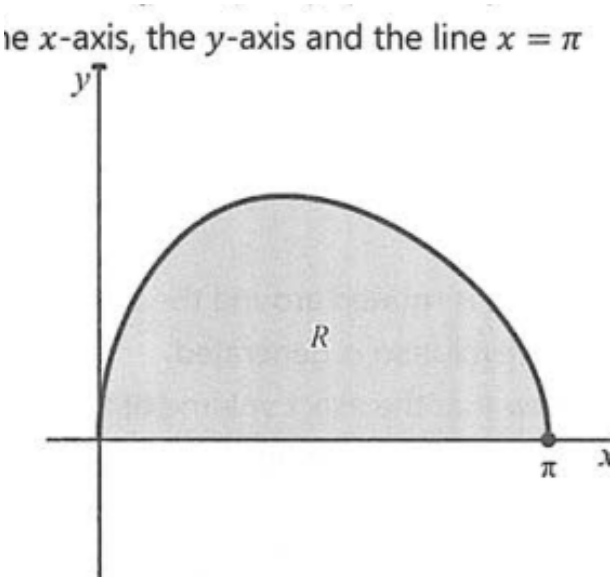
A region is bounded by the curve $y = \sin^2 x + 1$, the x-axis, the y-axis and the line $x = \pi$. The region is rotated through 2π radians about the x-axis to form a solid of revolution.

b) Find the exact volume of the solid formed.

Question 13.

a) Using the substitution $u = 2 - \cos x$, find the exact value of $\displaystyle\int_0^{\pi} \frac{\sin x}{\sqrt{2 - \cos x}}\, dx$.

A region R is bounded by the curve $y = 2(\sin x)^{1/2}(2 - \cos x)^{-1/4}$, the x-axis, the y-axis and the line $x = \pi$. The region is rotated through 2π radians about the x-axis.



b) Find the exact volume of the solid formed.

Question 14. A region is bounded by the curve $y = \sqrt{xe^x}$, the x-axis and the line $x = \ln k$ for constant $k > 0$. The region is rotated through 2π radians about the x-axis to form a solid of revolution of volume $\pi(81 \ln 3 - 20)$ units^3. Determine the value of k. **Prerequisite Knowledge for Question 15: Harmonic Form**

Question 15. The region R is bounded by the curve with equation

$$y = \frac{8}{2\cos x + \sqrt{5}\sin x}$$

the x-axis, the y-axis and $x = \frac{\pi}{4}$. Show that when the region R is rotated around the x-axis until a solid of revolution is generated, its volume is $32\pi(\sqrt{5}-2)$ units^3. **Prerequisite Knowledge for Question 16: Applications of De Moivre's Theorem**

Question 16.

a) Given that $z - \frac{1}{z} = 2i\sin\theta$, show that

$$\sin^6\theta = \frac{5}{16} - \frac{15}{32}\cos 2\theta + \frac{3}{16}\cos 4\theta - \frac{1}{32}\cos 6\theta.$$

A decorative glass ornament with a flat base is symmetrical about its horizontal axis. The curve of the glass ornament may be modelled by the curve with equation

$$y = H\sin^3\left(\frac{\pi x}{8}\right) \qquad \text{for } 0 \le x \le 4$$

where H is a positive constant and x is in centimetres.

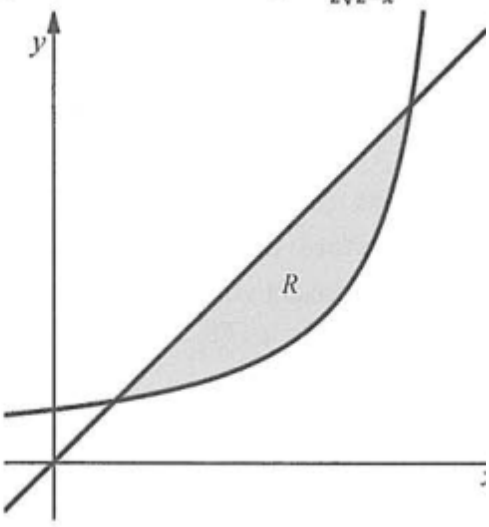
b) Given that the maximum height of the dome is 3 cm, write down the value of H.

c) Using algebraic integration and the result from part (a), determine the exact volume of the glass dome, in cm^3.

d) State one limitation of the model.

Question 17. The shaded region is bounded by the line $y = x$ and the curve $y = \dfrac{1}{2\sqrt{2} - x}$. The region is

line $y = x$ and the curve $y = \frac{1}{2\sqrt{2}-x}$



rotated around the x-axis until a solid of revolution is generated. Determine the volume of the solid generated giving your answer in exact form. **Prerequisite Knowledge for Question 18: Polar Curves**

Question 18. A curve is given by the polar equation $r = \dfrac{3}{2 - \cos\theta}$.

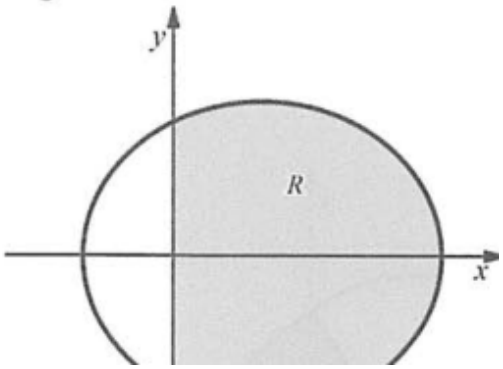
a) Show that the Cartesian equation of the curve can be written $y^2 = \dfrac{(x+3)^2}{4} - x^2$.

b) Explain why the curve is symmetrical in the x-axis.

c) A shaded region is bounded by the curve above, the x-axis and the y-axis as shown in the diagram below.

$y^2 = \frac{(x+3)^2}{4} - x^2$

A shaded region is bounded by the curve above, the x-axis and the y-axis as shown the diagram below.



The region is rotated around the x-axis until a solid of revolution is generated. Find the volume of the solid generated giving your answer in exact form.

Prerequisite Knowledge for Question 19: Polar Curves, A-Level Maths Partial Fractions and Reciprocal Trigonometric Functions

Question 19. A curve is given by polar equation $r = 3 - 2\sec\theta$.

a) Show that the Cartesian equation of the curve can be written in the form $y^2 = \dfrac{9x^2}{(x+2)^2} - x^2$.

b) Explain why the curve is symmetrical in the x-axis.

c) Show that $y^2 = 9 + \dfrac{36}{(x+2)^2} - \dfrac{36}{x+2} - x^2$.

A shaded region is bounded by the curve above, the x-axis, the y-axis and the line $x = 1$. The region is rotated around the x-axis until a solid of revolution is generated.

d) Show that the volume is given by $\pi\left(\frac{44}{3} + 36\ln\frac{2}{3}\right)$ units^3.

The following questions require the use of A-Level Further Maths Integration techniques

Question 20. For each of the following, the region is bounded by the x-axis, the y-axis and the given line. The region is rotated around the x-axis until a solid of revolution is generated. In each case, determine the volume of the solid generated giving your answer in exact form:

a) $y = (1 - x^2)^{-1/4}$ with line $x = \frac{1}{2}$

b) $y = \dfrac{3}{\sqrt{x^2+9}}$ with line $x = 4$

c) $y = (8 - 4x - 4x^2)^{-1/4}$ with line $x = \frac{1}{4}$

d) $y = \dfrac{1}{\sqrt{x^2+2}}$ with line $x = \sqrt{6}$

e) $y = \dfrac{1}{9+x^2}$ with line $x = 1$

Prerequisite Knowledge for Question 21: A-Level Maths Partial Fractions or Hyperbolic Functions

Question 21. A shaded region is bounded by the curve $y = \dfrac{1}{\sqrt{4-x^2}}$ for $0 \leq x < 1$, the x-axis and the y-axis. The region is rotated around the x-axis until a solid of revolution is generated. Determine the exact volume of the volume generated. **Prerequisite Knowledge for Question 22: Hyperbolic Functions**

Question 22. A shaded region is bounded by the curve $y = \dfrac{1}{(x^2+3)^{1/4}}$, the x-axis, the y-axis and the line $x = 2$. The region is rotated around the x-axis until a solid of revolution is generated. Show that the volume of the solid generated is given by

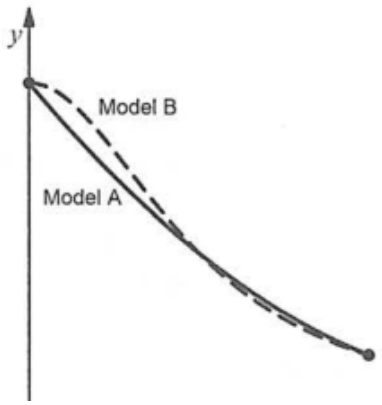
$$\pi \ln\left(\frac{p+\sqrt{q}}{\sqrt{r}}\right)$$

where p, q and r are integers to be found.

Question 23. A designer is creating a fountain spout which is 2 m long, with flat circular ends. The spout is symmetrical about its midpoint and should have diameters of 4 m at the top of the spout and 0.8 m at the bottom of the spout. The shape is modelled by rotating a curve about the x-axis. The origin O is at the top of the spout. Two possible equations for the curve are:

$$\text{Model A: } y = 0.2x^2 + ax + b \qquad \text{Model B: } y = \frac{p}{x^2 + q}$$

e spout and 0.8 m at the bottom of the
out.



for constants a, b, p and q.

a) Show that $a = -1.2$ and $b = 2$.
b) Find the values of p and q.
c) Use integration to find the volume predicted by model A giving your answer to three significant figures.
d) Use integration to find the volume predicted by model B giving your answer to three significant figures.

Prerequisite Knowledge for Question 24: Integration of Inverse Hyperbolic Functions

Question 24. A region R is bounded by the curve $y = \sqrt{\text{arcosh}(x)}$, the x-axis and the lines $x = 1$ and $x = 2$. The region is rotated through 2π radians about the x-axis to form a solid of revolution. Show that the exact volume formed is $\pi(2\ln(2 + \sqrt{3}) - \sqrt{3})$.

Question 25. A shaded region is bounded by the curve $y = \dfrac{1}{(k-x^2)^{1/4}}$, the x-axis, the y-axis and the line $x = k$ where k is a positive constant. The region is rotated around the x-axis until a solid of revolution is generated.

a) Determine the exact volume of the solid generated giving your answer in exact form in terms of k.

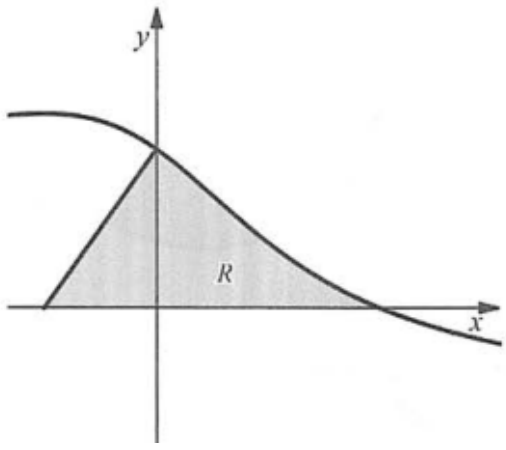
b) If the region lies entirely between $x = 0$ and $x = \frac{1}{2}$, find the volume.

Prerequisite Knowledge for Question 26: Calculus with Hyperbolic Functions

Question 26. The shaded region below is bounded by the line $y = \sqrt{2}x + \sqrt{2}$, the x-axis and the curve given by

$$y = -\frac{x-2}{\sqrt{x^2+2}}.$$

$\sqrt{x^2+2}$



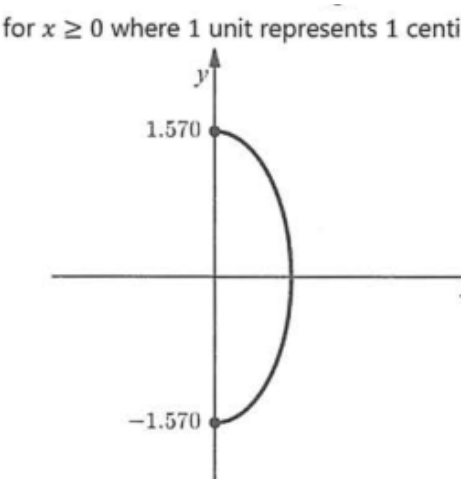
The region is rotated around the x-axis until a solid of revolution is generated. Determine the volume of the solid generated giving your answer to three significant figures.

Question 27. Find the exact volume created by rotating the curve through 2π radians around the y-axis for each of the following:

a) $x = (2y+1)^{1/3}$ for $0 \leq y \leq \frac{7}{2}$
b) $x = \operatorname{cosec} y$ for $\frac{\pi}{6} \leq y \leq \frac{\pi}{3}$
c) $x = \sqrt{\frac{2\sin y}{1+\cos y}}$ for $0 \leq y \leq \frac{\pi}{2}$
d) $x = \sqrt{y}\sec y$ for $0 \leq y \leq \frac{\pi}{4}$
e) $x = \sec y - \tan y$ for $0 \leq y \leq \frac{\pi}{4}$

Question 28. Zaynab is designing 50 small decorative beads which she is making out of resin. The cross-section of each bead is modelled by the curve

$$9x^2 + 2y^2 - \cos\left(\frac{\pi y}{3}\right) = 5$$



for $x \geq 0$ where 1 unit represents 1 centimetre. The curve is rotated about the y-axis through 2π radians to form the bead. The top and bottom of the bead correspond to the points $(0, 1.570)$ and $(0, -1.570)$ rounded to four significant figures.

a) Determine, according to the model, the volume of one bead.
b) Zaynab has 215 cm^3 of resin available. Use your answer to part (a) to decide whether she has enough material to make all 50 beads.

Question 29. The inner wall of a conical flask is modelled by $y = \ln x$ where $x \geq 1$. The region bounded by the curve, the y-axis and the lines $y = 0$ and $y = \ln 4$ is rotated about the y-axis. Determine the exact volume of the solid formed.

Question 30. A tent has a smooth, curved canopy. A vertical cross-section through the centre of the tent is modelled by

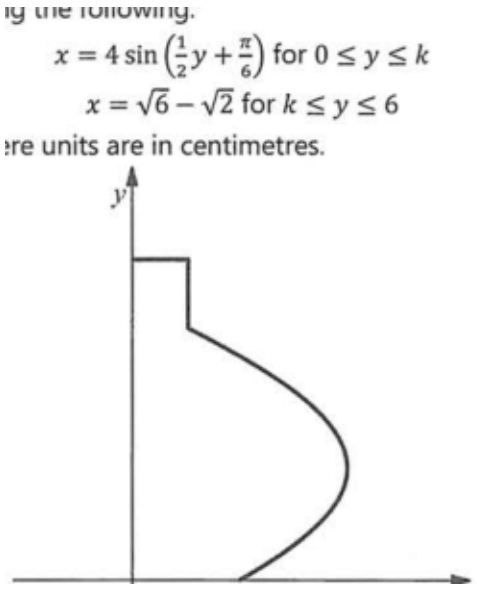
$$x^2 + \sin\left(\frac{y}{h}\right) = 1$$

for $0 \leq y \leq \frac{\pi h}{2}$, where x is the horizontal distance from the centre line in metres, y is the height above the ground in metres and h is the height of the tent. Given that this model finds the exact volume of air enclosed by the tent to be $(2\pi^2 - 4\pi)$ m^3, determine the height of the tent.

Question 31. The diagram below shows a cross-section of a design for a perfume bottle, which is formed by rotating it about the y-axis. The base of the bottle lies on the x-axis. It may be modelled using the following:

$$x = 4\sin\left(\frac{1}{2}y + \frac{\pi}{6}\right) \quad \text{for } 0 \le y \le k, \qquad x = \sqrt{6} - \sqrt{2} \quad \text{for } k \le y \le 6,$$

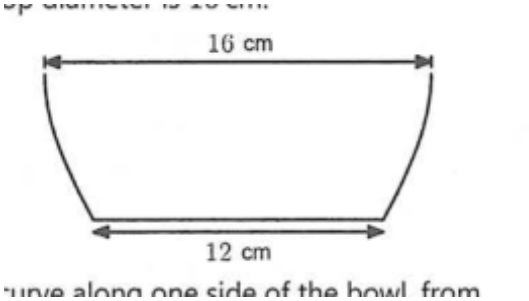
where units are in centimetres.



a) Show that the value of k is $\frac{3\pi}{2}$.
b) Find the diameter of the bottle at its base.
c) Determine the total volume of the perfume bottle giving your answer to three significant figures.
d) State one limitation of the model.

Prerequisite Knowledge for Question 32: Connected Rates of Change

Question 32. A bowl has a circular horizontal cross-section. The bottom diameter of the bowl is 12 cm and the top diameter is 16 cm.



The curve along one side of the bowl, from bottom to top, may be modelled by

$$x = k + 2\sin\left(\frac{\pi y}{12}\right)$$

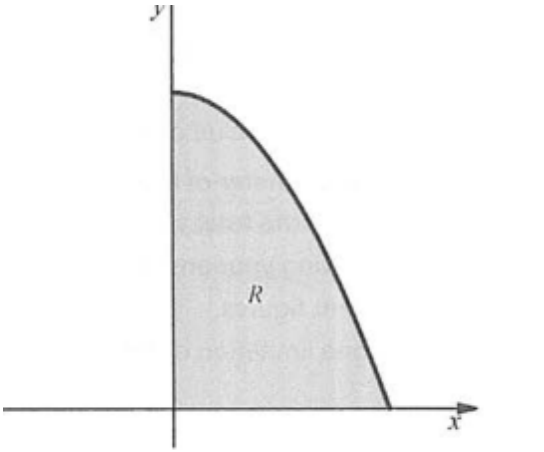
where x and y are in cm, k is a positive constant and $y = 0$ corresponds to the base of the vase.

a) Find the value of k.
b) Find the height of the bowl according to the model.

The bowl is being filled with water from a tap.

c) Find, in terms of h, the volume of water in the bowl when the water reaches a height of h cm.
d) If water flows in at a constant rate of 0.5 litres per minute, find the rate at which the water level is rising when the depth of water is 4 cm giving your answer to three significant figures.

Question 33. A designer is creating a solid paperweight. The central vertical cross-section of the paperweight is described by the curve $20x^2 + \ln(y+1) = 0.1$ for $y \geq 0$.



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A region R bounded by this curve and the y-axis is rotated through 2π radians about the y-axis to form the solid paperweight. Units are in metres.

a) Determine the volume of the paperweight giving your answer to three significant figures.
b) Given the paperweight is made from material of density 4.5 g cm^{-3}, determine its mass in kilograms.
c) Comment on the suitability of the model.

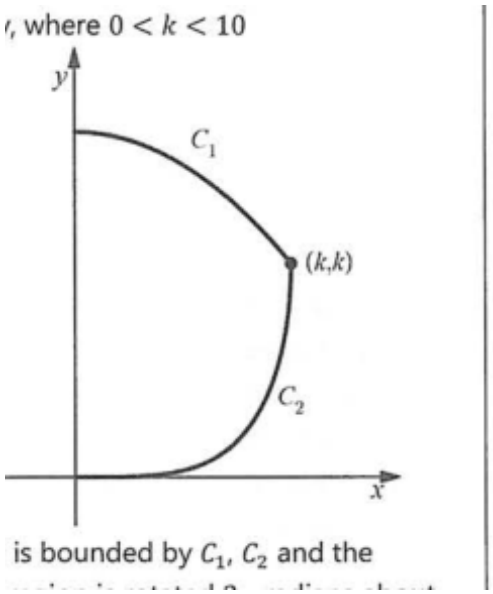
Prerequisite Knowledge for Question 34: Calculus of Hyperbolic Functions

Question 34. A region R is bounded by the curve $y = \sinh\left(\frac{1}{2}x^2\right)$, the x-axis, the y-axis and the line $y = \sinh 1$. The region is rotated around the y-axis through 2π radians. Show that the exact volume of the solid formed is $2\pi\left(1 - \frac{1}{e}\right)$.

Question 35. The curves C_1 and C_2 are given by equations

$$y = \frac{k-10}{k^2}x^2 + 10, \quad 0 \le x \le k, \qquad x = k\left(\frac{y}{k}\left(2 - \frac{y}{k}\right)\right)^{1/4}, \quad 0 \le y \le k,$$

respectively, where $0 < k < 10$.



A region R is bounded by C_1, C_2 and the y-axis. The region is rotated 2π radians about the y-axis. Show that the expression for the volume of the solid as a definite integral in terms of k is

$$\frac{1}{4}k^2\pi\big(k\pi - 2(k-10)\big) \text{ units}^3.$$

The following questions require the use of A-Level Further Maths Integration techniques

Question 36. The region R is bounded by the curve $y = \sqrt{\frac{1}{x^2} - 1}$ for $x > 0$, the x-axis, the y-axis and the line $y = 1$. The region R is rotated about the y-axis to form a solid of revolution. Find the exact volume of the solid formed.

Question 37. The equation of a curve is given by $x = \dfrac{1}{\sqrt[4]{k - y^2}}$ where k is a positive constant and $k > 1$. The region between the y-axis, the x-axis and the line $y = 1$ is rotated through 2π radians about the y-axis. Determine the exact volume generated giving your answer in terms of k. **Prerequisite Knowledge for Questions 38–39: Calculus with Hyperbolic Functions**

Question 38. The equation of a curve is given by $x = \dfrac{1}{\sqrt[4]{k + y^2}}$ where k is a positive constant. The region between the y-axis, the x-axis and the line $y = 1$ is rotated through 2π radians about the y-axis. Given that the volume of the solid of revolution formed is $\pi \ln\left(\frac{1+\sqrt{10}}{3}\right)$ units^3, find the exact value of k.

Question 39. A region R is bounded by the curve $y = \operatorname{arcosh} x$, the y-axis and the line $y = \frac{1}{2}\ln 5$. The line and the curve intersect at the point P.

a) Sketch the region R.

b) Find the exact coordinates of the point P.

c) The region is rotated through 2π radians about the y-axis to form a solid of revolution. Determine the exact volume of the solid formed.

Prerequisite Knowledge for Question 40: Linear Rational Functions

Question 40. It is given that $f(x) = \dfrac{ax+7}{x+b}$ where a and b are constants. It is known that $y = f(x)$ has asymptotes $x = -3$ and $y = 4$.

a) Write down the values of a and b.

b) Sketch $y = f(x)$.

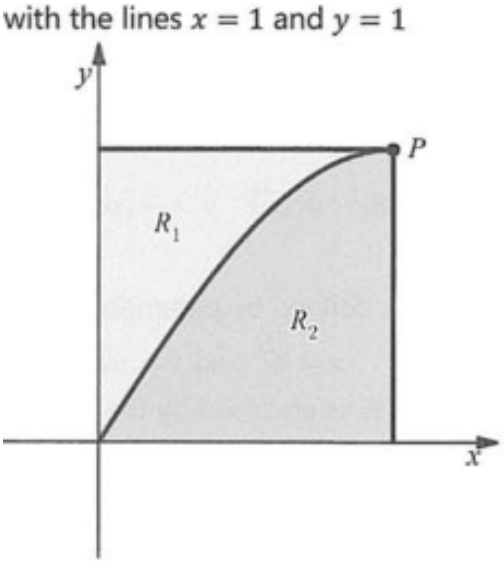
A finite region R is defined to be enclosed by the curve $y = f(x)$, the negative x-axis and the positive y-axis. Find the exact volume formed when rotating 2π radians around the:

c) x-axis

d) y-axis

Prerequisite Knowledge for Question 41: Further Maths Integration Methods

Question 41. The equation of a curve is given by $y = \sin\left(\frac{\pi x}{2}\right)$ where $0 \leq x \leq 1$. The point P is the point of intersection of the curve with the lines $x = 1$ and $y = 1$.



The region R_1 is bounded by the curve, the line $y = 1$ and the y-axis. The region R_2 is bounded by the curve, the line $x = 1$ and the x-axis.

a) Show that the exact volume of the solid formed when rotating region R_2 about the x-axis is $\frac{\pi}{2}$ units^3.

b) Show that the exact volume of the solid formed when rotating region R_1 about the y-axis is $\left(\pi - \frac{8}{\pi}\right)$ units^3.